We have an annuity that pays C dollars at the end of each year for t years. The annual interest rate is r

A payment received k years from today is worth:

$$\frac{C}{(1+r)^k} = C(1+r)^{-k}$$

PV of a single payment we will receive in year k

To find the discrete PV of this annuity, we sum up the PV of each individual payment:

$$PV_{discrete} = \sum_{k=1}^{n} C\cdot(1+r)^{-k}$$

#1

Now, let's compound interest more often

Let's say each year t has n subperiods (e.g. $n=4$ for quarters)

interest rate $r \longrightarrow \frac{r}{n}$ (e.g. 4%/yr = 1%/quarter)

cash flow $C \longrightarrow \frac{C}{n}$ (e.g. \$12,000/yr = \$3,000/quarter)

number of compounding periods $t \longrightarrow n\cdot t$ (e.g. at year 2, we've had $4\cdot 2 = 8$ quarters)

a year t between year 1 and the final year T

total compounding periods: nT

Thus, our PV becomes:

$$PV_n = \sum_{k=1}^{nT} \frac{C}{n}\left(1+\frac{r}{n}\right)^{-k}$$

compounded over n periods

#2

So, where does e^{-rt} come from? Why e, and why not $(1+r)$?

The discount factor for a pmt in sub-period $k = nt$ is:

$$\left(1+\frac{r}{n}\right)^{-nt}$$

by the defining limit of e: (A)

$$\lim_{n\to\infty}\left(1+\frac{r}{n}\right)^{n} = e^{r}$$

so

$$\lim_{n\to\infty}\left(1+\frac{r}{n}\right)^{-nt}$$

$$= \left[\left(1+\frac{r}{n}\right)^{n}\right]^{-t} \xrightarrow[n\to\infty]{} e^{-rt}$$

#3

(A) What's the defining limit of e & how is it used like this?

definition of e: $e = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^{n}$

let $m = \frac{n}{r}$

so $rm = n$

& $\frac{r}{n} = \frac{1}{m}$

we want

$$\lim_{n\to\infty}\left(1+\frac{r}{n}\right)^{n} = \left(1+\frac{1}{m}\right)^{rm}$$

$$\lim_{n\to\infty}\left[\left(1+\frac{1}{m}\right)^{m}\right]^{r} = e^{r}$$

this is e.

As $n\to\infty$, $m\to\infty$